ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

DISTANCE LEARNING PROGRAMME
(Academic Session : 2024 - 2025)

JEE (Advanced)
TEST # 13
22-12-2024

JEE(Main+Advanced) : LEADER TEST SERIES / JOINT PACKAGE COURSE

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A	A	A	D	B	D	B	B
SECTION-I (ii)	Q.	9	10	11	12	13	14		
	A.	A,B	A,C,D	B,C,D	B,C,D	A,B,C,D	A,D		
SECTION-III	Q.	1	2	3	4	5	6		
	A.	5	9	6	7	5	4		

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	C	D	B	D	C	C	A	B
SECTION-I (ii)	Q.	9	10	11	12	13	14		
	A.	A,B,C,D	A,B,C	A,B	A,C	A,B,D	A,B,C		
SECTION-III	Q.	1	2	3	4	5	6		
	A.	2	7	4	4	8	4		

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	D	C	D	A	B	C	D	A
SECTION-I (ii)	Q.	9	10	11	12	13	14		
	A.	A,D	A,D	B,C	A	A,B,C	A,B		
SECTION-III	Q.	1	2	3	4	5	6		
	A.	7	9	4	4	6	6		

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A)

Let piston be displaced 'x' towards left than volume decreases and pressure increases

PV = constant {As isothermal}

Let P, & V are initial pressure & volume

$PV = (P + \Delta P)(V - \Delta V)$

{ΔP is increase in pressure}

$PV = PV + \Delta PV - P\Delta V - \Delta P\Delta V$

[neglect $\Delta P \cdot \Delta N$]

$\Rightarrow \Delta PV = P\Delta V$

$\Delta P\,(Ah) = PAx \Rightarrow \Delta P = P\frac{x}{h}$

$F = \Delta P \cdot A = PA\frac{x}{h} = -Ma \qquad [a = -\omega^2 x]$

By comparing

$\Rightarrow \omega = \sqrt{\frac{PA}{Mh}} \Rightarrow T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{Mh}{PA}}$

Method-2

Dimension analysis can be done

2. Ans (A)

$\frac{6}{\cos\theta} = \frac{4}{\cos 60^\circ}$

$\cos\theta = \frac{3}{4}$

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3. **Ans (A)**

As cork move from bottom to top it reach to its terminal speed.

4. **Ans (D)**

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

$$\frac{1}{\lambda_1} = R\left(\frac{1}{3^2} - \frac{1}{4^2}\right)$$

$$\frac{1}{\lambda_1} = R\left(\frac{7}{9\times 16}\right)$$

$$\frac{1}{\lambda_2} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = R\left(\frac{5}{4\times 9}\right)$$

$$\frac{\lambda_1}{\lambda_2} = \frac{\frac{5}{36}}{\frac{7}{9\times 16}} = \frac{20}{7}$$

5. **Ans (B)**

$A_n = \pi r^2 \qquad \Rightarrow A_n = \pi\left(r_r n^2\right)^2$

$A_n = \pi r_0^2 n^4 \qquad$ so, $\ell n\frac{A_n}{A_1} = 4\ell n\ n$

6. **Ans (D)**

$$\frac{\Delta g}{g} = \frac{\Delta L}{L} + 2\frac{\Delta T}{T}$$

In option (D) error in Dg is minimum and number of repetition of measurement are maximum. In this case the error in g is minimum.

7. **Ans (B)**

The potential difference across zener is 3 volt.

$\therefore$ Current through 100Ω is $= \frac{3}{100} = 0.03A$

The potential difference across 20Ω is 2 volt

$\therefore$ Current through 20Ω is $= \frac{2}{20} = 0.1A$

Current through zener is $= 0.1 - 0.03 = 0.07$

and hence the power $= V_Z \times I_Z = 3 \times .07 = 0.21$ W

8. **Ans (B)**

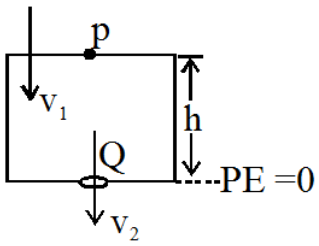
$\overline{A + B} = \bar{A}.\bar{B}$

PART-1 : PHYSICS

SECTION-I (ii)

9. **Ans (A,B)**

Applying Bernoulli's equation between P and Q



$$p_0 + \rho gh + \frac{1}{2}\rho v_1^2 = \rho_0 + 0 + \frac{1}{2}\rho v_2^2$$

$$\rho gh = \frac{\rho}{2}\left(v_2^2 - v_1^2\right)$$

$v_2^2 - v_1^2 = 2gh$ .....(i)

From equation of continuity,

$$Av_1 = av_2 = \frac{A}{3}v_2$$

$\Rightarrow v_2 = 3v_1$

$9v_1^2 - v_1^2 = 2gh$

$8v_1^2 = 2gh$

$$\Rightarrow v_1 = \sqrt{\frac{gh}{4}} = \frac{\sqrt{gh}}{2}$$

Acceleration of top layer

$$a = \frac{dv_1}{dt} = \frac{1}{2}\sqrt{g} \times \frac{1}{2\sqrt{h}} \times \frac{dh}{dt}$$

$$= \frac{1}{4}\sqrt{\frac{g}{h}} \times (--v_1) = -\frac{1}{4}\sqrt{\frac{g}{h}} \times \frac{\sqrt{gh}}{2} = -\frac{g}{8}\text{m/s}^2$$

10. **Ans (A,C,D)**

(A) $E_K - E_L + E_L - E_N = E_K - E_W = E_K$. K_r ray

$= E_K - E_M + E_M - E_N$

11. **Ans (B,C,D)**

$$u = \sqrt{\frac{2\times 10}{1200}[1200\times 0.1 + 600\times 0.3]}$$

$v = \sqrt{5}$

$$F = \rho av^2 = 1200 \times \frac{5}{10^4} \times 5 = 3$$

$f_{max} = \mu N$

$$f_{max} = 0.04\left[\frac{600\times 30}{100} + \frac{1200\times 20}{100}\right] \times 1000 \times 10^{-4} \times 10$$

$f_{max} = 16.8N$

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12. Ans (B,C,D)

$\because a = -\omega^2 y$ and at t = T, y is maximum so

acceleration is maximum at t = T.

Also y=0 at $t = \frac{3T}{4}$, so force is zero at $t = \frac{3T}{4}$

At $t = \frac{T}{2}$, v=0 $\Rightarrow$ PE = oscillation energy

13. Ans (A,B,C,D)

λ_2 remains same

λ_1 may increase, decrease or even be same

14. Ans (A,D)

$u_s = u_i$

$\frac{2h}{\lambda} = 2mu$

$E_i = \frac{1}{2}mu^2 = \frac{h^2}{2m\lambda^2}$

$\lambda = \frac{c}{f} \Rightarrow E_i = \frac{h^2 f^2}{2mc^2}$

PART-1 : PHYSICS

SECTION-III

1. Ans (5)

From phaser diagram the total angle covered by

block is 90° + 30° +30° + 90° = 240°

time taken = $\frac{240}{360}T = \frac{2T}{3}$

m = 2 , n = 3

2. Ans (9)

$^{12}_{5}B \rightarrow {}^{12}_{6}C + {}^{0}_{-1}e + \bar{\nu}$

Mass defect = (12.014 − 12) u

∴ Released energy = 13.041 MeV

Energy used for excitation of $^{12}_{6}C$ = 4.041 MeV

∴ Energy converted to KE of electron

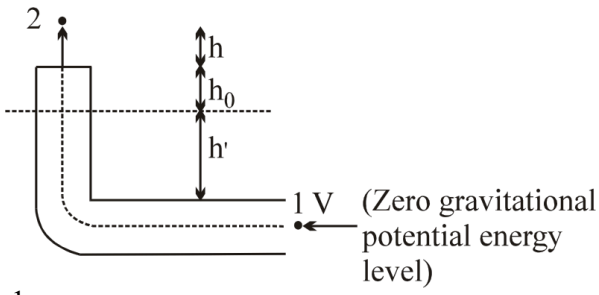
= 13.041 − 4.041 = 9 MeV

3. Ans (6)

Applying Bernoulis theorem for points 1 and 2

of the streamline shown

$P_0 \rightarrow$ Atmospheric pressure



$\frac{1}{2}\rho V^2 + (P_0 + \rho g h') + 0$

$= 0 + P_0 + \rho g(h' + h_0 + h)$

$\Rightarrow$ h = 60 cm

4. Ans (7)

$T = \pi\sqrt{\frac{5}{80}} + \pi\sqrt{\frac{5}{180}}$

$= \pi\sqrt{\frac{1}{16}} + \pi\sqrt{\frac{1}{36}} = \frac{\pi}{4} + \frac{\pi}{6}$

$\Rightarrow \frac{5\pi}{12} = \frac{p}{q}\pi$

5. Ans (5)

$K\varepsilon_\alpha = \frac{Q \times m_y}{m_\alpha + m_y}$

$48 = \frac{50 m_y}{4 + m_y}$

$m_y = 96$

∴ mass of parent nuclei $= m_\alpha + m_y = 100$

6. Ans (4)

$E(t) = A^2 e^{-\alpha t}$

$\ln E = \ln A^2 + \ln e^{-\alpha t}$

$\ln E = 2 \ln a - \alpha t$

$\frac{\Delta E}{E} \times 100 = 2\frac{\Delta A}{A} \times 100 + \alpha \Delta t \times 100$

$= 2 \times 1.25 + \frac{2}{10} \times 1.5 \times 5$

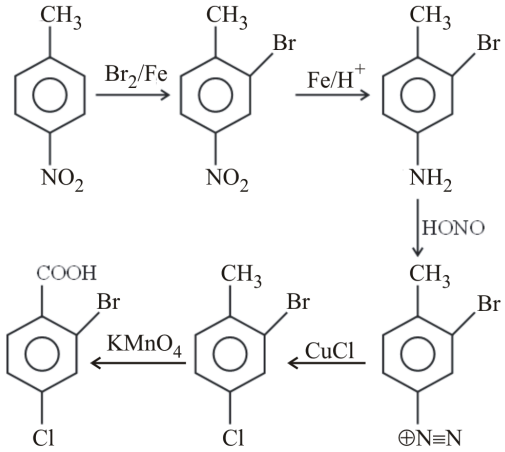
$= 2.5 + 1.5 = 4\%$

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PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (C)

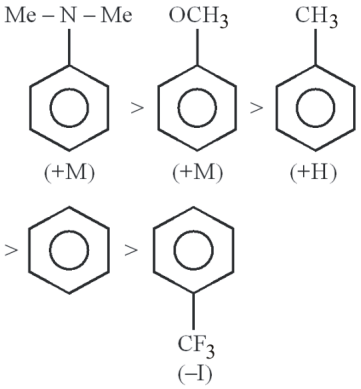


2. Ans (D)

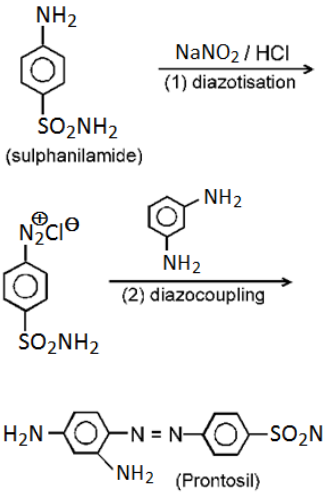
2° amines form N-nitrosoamine upon treatment with $NaNO_2$ & HCl.

3. Ans (B)

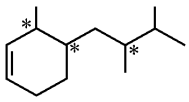
Higher the electron density on Benzene Ring, Higher its Reactivity towards electrophilic substitution Reaction



4. Ans (D)



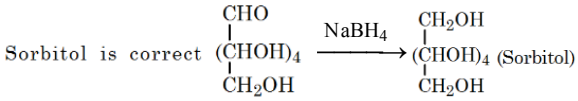
5. Ans (C)



6. Ans (C)

Milk sugar (lactose) C1 - C4 β-glycosidic linkage is present.

7. Ans (A)



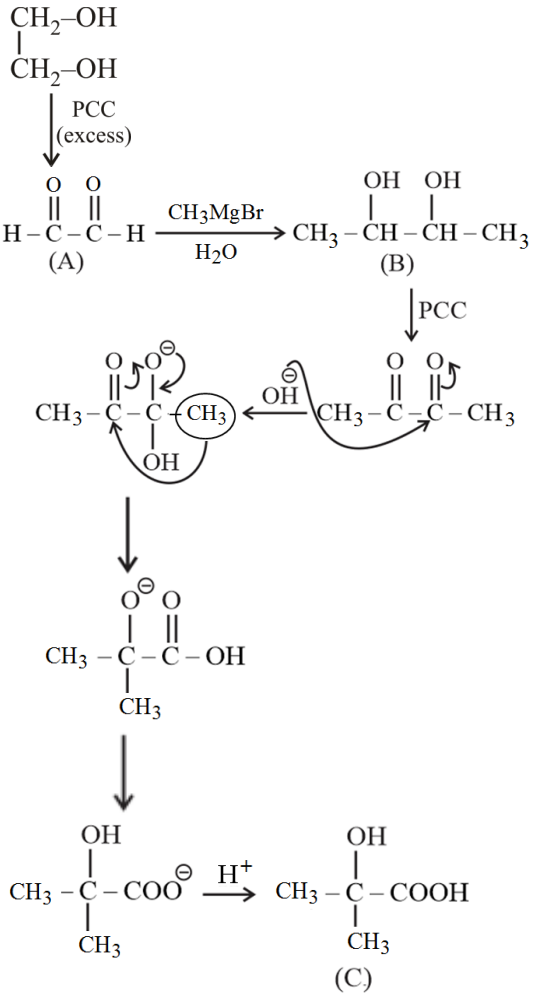
8. Ans (B)

In paper chromatography, a special quality paper known as chromatography paper is used. Paper contains water trapped in it, which acts as the stationary phase.

PART-2 : CHEMISTRY

SECTION-I (ii)

9. Ans (A,B,C,D)



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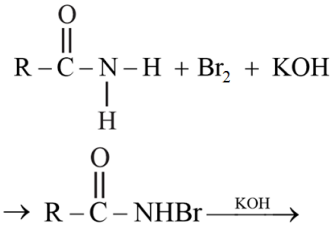
10. **Ans (A,B,C)**

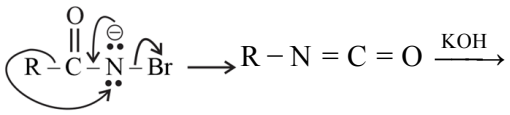
P and Q are geometrical isomer

R and S are metamer

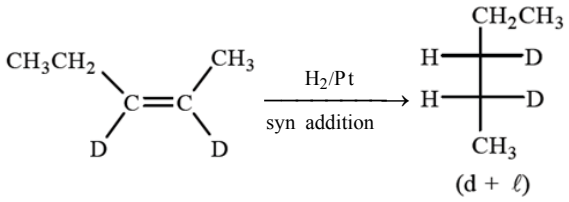
P to R is known as Beckmann rearrangment

11. **Ans (A,B)**





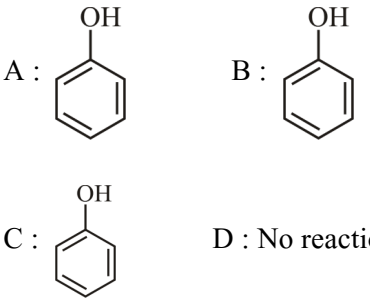
12. **Ans (A,C)**



13. **Ans (A,B,D)**

Hemiacetal group present in Compound (A), (B), and (D)

14. **Ans (A,B,C)**



PART-2 : CHEMISTRY

SECTION-III

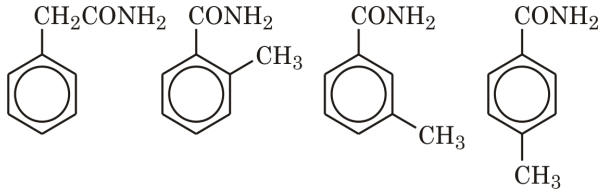
1. **Ans (2)**

(i), (ii),

2. **Ans (7)**

Carbyl amine reaction gives foul smell with 1° amine and aniline.

3. **Ans (4)**

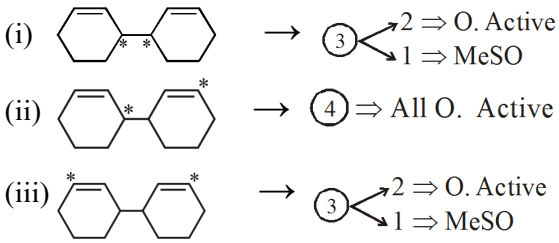


4. **Ans (4)**

Conceptual

5. **Ans (8)**

Possible Product



Possible total O.Active Product = 8

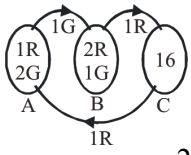
6. **Ans (4)**

(i, ii, v, vi)

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (D)**



Probability = $\frac{2}{3} \times \frac{2}{4} \times \frac{1}{2} = \frac{1}{6}$

2. **Ans (C)**

$f'(x) = 3x^2 + 2ax + 9$ .....(i)

for $f(x)$ to be invertible it must be a monotonic function because it is onto always.

$\Rightarrow f'(x) \geq 0 \ \forall \ x$

$\Rightarrow D \leq 0$ for (i)

$\Rightarrow a^2 - 3b \leq 0$

This is true only for 16 ordered pair

(a, b) {a, b Î [1, 6]}

$\therefore$ required probability = $\frac{16}{36} = \frac{4}{9}$

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3. Ans (D)

$P(A'B) + P(AB') = P(B'C) + P(BC') = P(A'C) + P(AC') = \frac{1}{4}$

$P(ABC) = \frac{1}{16}$

$P(A \cup B \cup C) = P(A) + P(B) + P(C) - (P(AB) + P(BC) + P(CA) + P(ABC))$

$P(A'B) = P(AB') = P(A) + P(B) - 2P(AB)$

$P(A'B) + P(AB') + P(B'C) + P(BC') + P(A'C) + P(AC')$

$= 2(P(A) + P(B) + P(C) - P(AB) - P(AC) - P(AC))$

$P(A) + P(B) + P(C) - +(P(AB) + P(BC) + P(AC) = \frac{3}{8}$

$\therefore\ P(A \cup B \cup C) = \frac{3}{8} + \frac{1}{16} = \frac{7}{16}$

4. Ans (A)

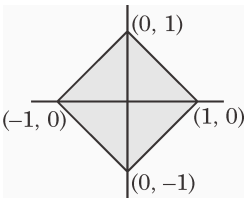
One-one mapping = n!

Total mapping = n^n

$\Rightarrow$ P(one-one) $= 1 - \frac{29}{32}$ $\Rightarrow \frac{n!}{n^n} = \frac{3}{32}$

$\Rightarrow$ n = 4 (By hit & trial)

5. Ans (B)



Now check the alternatives.

6. Ans (C)

Data 1	Data 2
$x = 5$	$x = 5$
$\bar{x} = 2$	$\bar{y} = 4$
$\Rightarrow \Sigma x_i = 2 \times 5 = 10$	$\Rightarrow \Sigma y_i = 4 \times 5 = 20$
$var_x = 4$	$var_y = 5$
$\frac{\Sigma x_i^2}{5} - (2)^2 = 4$	$\frac{\Sigma y_i^2}{5} - (4)^2 = 5$
$\Sigma x_i^2 = 40$	$\Sigma y_i^2 = 105$

Combined variance $= \frac{\Sigma x_i^2 + \Sigma y_i^2}{10} - \frac{\Sigma x_i + \Sigma y_i}{10}$

$= \frac{40 + 105}{10} - \left(\frac{10 + 20}{10}\right)^2$

$= 5.5$

7. Ans (D)

Obvious

8. Ans (A)

$n(S) = {}^{10}C_3$

n(A) = No. of 4A . 2B & 1 C has been taken by first 7 contestant

$= {}^5C_4 \times {}^3C_2 \times {}^2C_1$

$P(A) = \frac{{}^5C_4 \times {}^3C_2 \times {}^2C_1}{{}^{10}C_3}$

$= \frac{5 \cdot 3 \cdot 2}{120} = \frac{1}{4}$

Option (A) is correct.

PART-3 : MATHEMATICS

SECTION-I (ii)

9. Ans (A,D)

Let A_i : Number i is selected from A.

B_j : Number j is selected from B.

$\Rightarrow$ $p = P(n(A) \le n(B)) = 1 - P(n(A) > n(B))$

$= 1 - \{P(A_2).P(B_1) + P(A_3).P(B_1 \text{ or } B_2)\}$

$= 1 - \left\{\frac{1}{3} \times \frac{1}{5} + \frac{1}{3} \times \frac{2}{5}\right\} = \frac{4}{5} = P(n(B) \le n(A))$

$= P(A_1).P(B_1) + P(A_2).P(B_1 \text{ or } B_2) + P(A_3)P(B_1$ or B_2 or $B_3)$

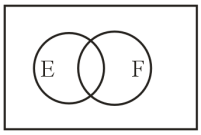
$= \frac{1}{3} \times \frac{1}{5} + \frac{1}{3} \times \frac{2}{5} + \frac{1}{3} \times \frac{3}{5}$

$q = \frac{2}{5}$

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10. Ans (A,D)

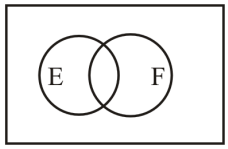
Let E and F are 2 events which can expressed on van diagram as follow:



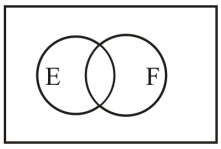
$$(A)\ P(E/F)+P(\bar{E}/F)=\frac{P(E\cap F)}{P(F)}+\frac{P(\bar{E}\cap F)}{P(F)}$$

$$=\frac{P(E\cap F)+P(\bar{E}\cap F)}{P(F)}$$

$P(E\cap F)=$ ($E\cap F$ is the shaded region)



$P(\bar{E}\cap F)=$



So $P(E\cap F)+P(\bar{E}\cap F)$ will be equal to PCF)

So $P(E/F)+P(E/F)=\frac{P(F)}{P(F)}=1$ (option A is true)

$$(B)\ P(E/F)+P(E/\bar{F})=\frac{P(E\cap F)}{P(F)}+\frac{P(E\cap \bar{F})}{P(F)}$$

$$=\frac{P(E\cap F)}{P(F)}+\frac{P(E\cap \bar{F})}{1-P(F)}\neq 1$$

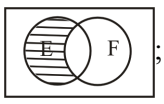
$$(C)\ P(\bar{E}/F)+P(E/\bar{F})$$

$$=\frac{P(\bar{E}\cap F)}{P(F)}+\frac{P(E\cap \bar{F})}{P(\bar{F})}=\frac{P(\bar{E}\cap F)}{P(F)}+\frac{P(E\cap \bar{F})}{1-P(F)}\neq 1$$

$$(D)\ P(E/\bar{F})+P(\bar{E}/\bar{F})$$

$$=\frac{P(E\cap \bar{F})}{P(\bar{F})}+\frac{P(\bar{E}\cap \bar{F})}{P(\bar{F})}=\frac{P(E\cap \bar{F})+P(\bar{E}\cap \bar{F})}{P(\bar{F})}$$

$P(E\cap \bar{F})=$



;

$P(\bar{E}\cap \bar{F})=$



So $P(E\cap \bar{F})+P(\bar{E}\cap \bar{F})=P(\bar{F})$

$$\therefore P\left(\frac{E}{\bar{F}}\right)+PP\left(\frac{\bar{E}}{\bar{F}}\right)=\frac{P(\bar{F})}{P(\bar{F})}=1$$ (D option is true)

11. Ans (B,C)

Ball	Balls composition	$P(B_i)$
B_1	$5R+5G$	$\frac{3}{10}$
B_2	$3R+5G$	$\frac{3}{10}$
B_3	$5R+3G$	$\frac{4}{10}$

$$(A)\ P(B_3\cap G)=P\left(\frac{G_1}{B_3}\right)P(B_3)$$

$$=\frac{3}{8}\times\frac{4}{10}=\frac{3}{20}$$

$$(B)\ P(G)=P\left(\frac{G_1}{B_1}\right)P(B_1)+P\left(\frac{G}{B_2}\right)P(B_2)+P\left(\frac{G}{B_3}\right)P(B_3)$$

$$=\frac{3}{20}+\frac{3}{16}+\frac{3}{20}=\frac{39}{80}$$

$$(C)\ P\left(\frac{G}{B_3}\right)=\frac{3}{8}$$

$$(D)\ P\left(\frac{B_3}{G}\right)=\frac{P(G\cap B_3)}{P(G)}=\frac{3/20}{39/80}=\frac{4}{13}$$

12. Ans (A)

P(Problem is solved by at least one of them)

= 1 – P(solved by none)

$$=1-\left(\frac{1}{2}\times\frac{1}{4}\times\frac{3}{4}\times\frac{7}{8}\right)$$

$$=1-\frac{21}{256}=\frac{235}{256}$$

13. Ans (A,B,C)

P(S) = P(A and (B or C))

$$\frac{1}{2}=P(A)\ .\ P(B\text{ or }C)$$

$$\frac{1}{2}=p\frac{1}{2}\left(q+\frac{1}{2}\right)$$

$$1=p\left(q+\frac{1}{2}\right)$$

So option (D) satisfies only

So A, B & C are correct.

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14. Ans (A,B)

(i) $P(A) = \frac{1}{5}$ $P(B) = \frac{7}{25}$ $P(B/A) = \frac{9}{10}$

$P(\bar{A} \cap \bar{B}) = 1 - P(A \cup B)$

$= 1 - [P(A) + P(B) - P(A \cap B)]$

$= 1 - \left[\frac{1}{5} + \frac{7}{25} - P(A) \cdot P(B/A)\right]$

$= 1 - \left[\frac{1}{5} + \frac{7}{25} - \frac{1}{5} \cdot \frac{9}{10}\right]$

(ii) $P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A) \cdot P(B/A)}{P(B)}$

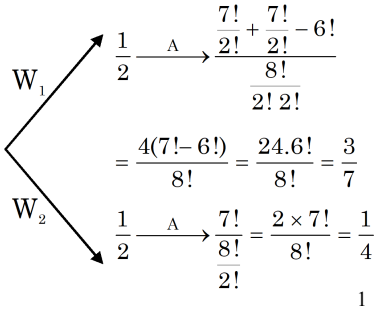
$= \frac{\frac{1}{5} \cdot \frac{9}{10}}{\frac{7}{25}} = \frac{9}{14} = \frac{18}{28}$

Option (AB) is correct.

PART-3 : MATHEMATICS

SECTION-III

1. Ans (7)



Required probability $= \frac{\frac{1}{2} \times \frac{3}{7}}{\frac{1}{2} \times \frac{3}{7} + \frac{1}{2} \times \frac{1}{4}}$

$= \frac{12}{19} = \frac{p}{q}$

$\therefore q - p = 7$

2. Ans (9)

If the sum of digits in a number is divisible by 9, then number itself is divisible by 9

$\Rightarrow$ either number comprises {1,2,4,5,6} or {1,2,3,5,7}.

$\Rightarrow$ required probability

$= \frac{2 \times 5!}{{}^7P_5} = \frac{2}{21} \Rightarrow \frac{x}{y} = \frac{2}{21} \Rightarrow y - 6x = 9$

3. Ans (4)

$x^3 + ax^2 + bx + C$

$a, b, c \in N \ (N < 10)$

{1, 2, 3, 4, 5, 6, 7, 8, 9}

Total number of polynomial functions $\longrightarrow$ 9.9.9

If function is divisible by $x^2 + 1$

then $f(i) = 0$ & $f(-i) = 0$

$f(i) = -i - a + bi + c = 0$

$(c - a) + (b - 1)i = 0$

$a = c$ & $b = 1$

$f(-i) = i - a - bi + c = 0$

$(c - a) - (b - 1)i = 0$

$a = c$ & $b = 1$

b is fixed hence a = c may take 9 values.

Probability $= \frac{9}{9.9.9} = \frac{1}{81}$

$\mu = \frac{1}{81}$

$\left(\frac{1}{3}\right)^{\log_{(1/81)}256} = 3^{\log_{(81)}256} = 4$

4. Ans (4)

Possible points that Humpty Not scores are

$= \begin{cases} 9 & \text{i.e. } 333 \\ 8 & \text{i.e. } 233 \\ 7 & \text{i.e. } 223 \end{cases}$

Required probability = 1–P(he score 7,8 or 9)

$= 1 - \{{}^3C_2(0.3)^2.(0.5) + {}^3C_2.(0.5)^2.(0.3) + {}^3C_3.(0.5)^3\}$

$= 1 - \left\{\frac{3.9.5 + 3.25.3 + 1.125}{1000}\right\} = \frac{103}{200}$

5. Ans (6)

$\frac{3 + 7 + 9 + 12 + 13 + 20 + x + y}{8} = 10$

$x + y = 16$

$\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2 = 25$

$3^2 + 7^2 + 9^2 + 12^2 + 13^2 + 20^2 + x^2 + y^2 = 1000$

$x^2 + y^2 = 148$

$xy = 54$

6. Ans (6)

First three prime numbers are 2,3 and 5.

for real roots $D > 0 \Rightarrow b^2 - 4ac > 0$

so $b \neq 2, b \neq 3 \Rightarrow b = 5$

$\therefore$ $b = 5$, $a = 2$, $b = 3$

& $b = 5$, $a = 3$, $b = 2$ i.e. in two ways.

Total number of ways of choosing a,bc

$= 3 \times 2 \times 1 = 6$

$\therefore$ required probability

$= \frac{2}{6} = \frac{1}{3} = P \Rightarrow 18P = 6$

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